

Multi-Model Architectural Benchmark: Stratified Held-Out Validation

(stratified_held_out_v2.json)

Evaluation Scope: 20 Subjects (40 Paired 3D Volumes: OD & OS) across 5 Architecture Checkpoints **Pipeline Configuration:** Continuous Surface Clamping (2.0 px axial tolerance) + Optic Cup-Reach Tracking (disc_cut_mode='cup') **Evaluation Protocol:** Stratified Held-Out V2 Benchmark (Frozen 2026-10-02; Evaluated 2026-10-08)

Stratified Benchmark Split Architecture: This report benchmarks 5 models over the full **20-subject (40-scan)** held-out cohort (stratified_held_out_v2.json). - **Human-Audited Corrected Tier:** 6 Subjects (12 scans: BEH0174, BEH0310, BEH0314, BEH0335, BEH0354, BEH0352) with expert manual corrections. - **Machine-Verified Tier:** 14 Subjects (28 scans: BEH0090, BEH0284, BEH0259, BEH0084, BEH0264, BEH0343, BEH0303, BEH0030, BEH0043, BEH0290, BEH0249, BEH0279, BEH0289, BEH0297) unedited commercial segmentations. - **Mutual Core Unseen Set:** BEH0314 and BEH0335 were strictly held out across **all baseline and expanded models**.

1. Executive Model Architecture Overview

Model ID	Architectural Description	Parameters	Training Cohort Scope	Biplanar Fusion	Status on V2 Split
18043443	2.5D ResNet Light (1.65M)	~1.65M	23-Subject Baseline	No (Unilateral)	Evaluated on V2 Cohort
18045386	2.5D ResNet Heavy (6.58M)	~6.58M	23-Subject Baseline	No (Unilateral)	Evaluated on V2 Cohort
18223981	Biplanar Orthogonal 23-Subj (6.58M)	~6.58M	23-Subject Baseline	Yes (H + V)	Evaluated on V2 Cohort
18266075	Biplanar Orthogonal 61-Subj (6.58M)	~6.58M	61-Subject Expanded	Yes (H + V)	Evaluated on V2 Cohort
18563914	Biplanar Orthogonal V2 Held-Out (6.58M)	~6.58M	Stratified V2 Protocol	Yes (H + V)	Native Held-Out (Strict)

2. Multi-Model Aggregate Clinical Benchmarks (All 40 Scans)

Model Identifier	Median RNFL Dice	Mean Dice ± SD	Median MABE	Mean MABE ± SD	Median \$P_{.95}\$	Median Cup IoU
2.5D ResNet Light (1.65M)	0.8055	0.7662 ± 0.1114	12.58 µm	20.18 ± 19.52 µm	32.96 µm	0.8422
2.5D ResNet Heavy (6.58M)	0.8562	0.8212 ± 0.1126	12.59 µm	19.33 ± 20.84 µm	34.48 µm	0.8109
Biplanar Orthogonal 23-Subj (6.58M)	0.9422	0.9260 ± 0.0781	4.91 µm	8.98 ± 16.69 µm	16.90 µm	0.9213
Biplanar Orthogonal 61-Subj (6.58M)	0.9018	0.9000 ± 0.0421	2.81 µm	3.90 ± 2.42 µm	7.15 µm	0.9444
Biplanar Orthogonal V2 Held-Out (6.58M)	0.9518	0.9441 ± 0.0305	4.71 µm	6.22 ± 5.84 µm	16.85 µm	0.9406

3. Stratified Subgroup Benchmarks

3.1 Human-Audited Corrected Tier (6 Subjects / 12 Scans)

Scored strictly against expert human manual boundary corrections.

Model Identifier	Median RNFL Dice	Mean Dice ± SD	Median MABE	Mean MABE ± SD	Median \$P_{.95}\$	Median Cup IoU
Light (1.65M)	0.8223	0.7664 ± 0.1108	11.38 µm	19.63 ± 16.58 µm	29.93 µm	0.8848
Heavy (6.58M)	0.8744	0.8466 ± 0.0675	10.59 µm	14.95 ± 10.15 µm	29.34 µm	0.8981
Biplanar 23s	0.9537	0.9369 ± 0.0354	4.23 µm	6.35 ± 4.91 µm	13.50 µm	0.9265
Biplanar 61s Exp	0.9454	0.9314 ± 0.0293	4.87 µm	5.92 ± 3.37 µm	15.75 µm	0.9365
Biplanar V2 (Latest)	0.9502	0.9438 ± 0.0259	4.73 µm	5.92 ± 3.25 µm	17.77 µm	0.9438

3.2 Machine-Verified Tier (14 Subjects / 28 Scans)

Model Identifier	Median RNFL Dice	Mean Dice ± SD	Median MABE	Mean MABE ± SD	Median \$P_{.95}\$	Median Cup IoU
Light (1.65M)	0.7986	0.7662 ± 0.1117	13.01 µm	20.42 ± 20.64 µm	33.59 µm	0.8230
Heavy (6.58M)	0.8534	0.8104 ± 0.1255	12.98 µm	21.20 ± 23.76 µm	36.36 µm	0.7895
Biplanar 23s	0.9396	0.9213 ± 0.0901	4.93 µm	10.11 ± 19.58 µm	17.22 µm	0.9172
Biplanar 61s Exp	0.8890	0.8865 ± 0.0395	2.66 µm	3.04 ± 1.01 µm	6.74 µm	0.9497
Biplanar V2 (Latest)	0.9524	0.9443 ± 0.0323	4.71 µm	6.34 ± 6.64 µm	16.68 µm	0.9398

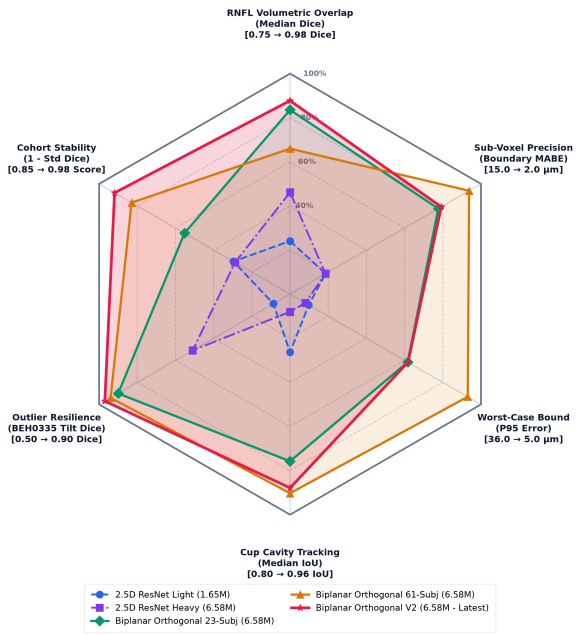
3.3 Mutual Core Unseen Outliers (BEH0314 & BEH0335: 4 Scans)

Model Identifier	Median RNFL Dice	Mean Dice ± SD	Median MABE	Median \$P_{.95}\$	Median Cup IoU
Light (1.65M)	0.6932	0.6715 ± 0.1426	28.06 µm	62.32 µm	0.5874
Heavy (6.58M)	0.8028	0.7910 ± 0.0901	17.22 µm	55.87 µm	0.7375
Biplanar 23s	0.9029	0.9039 ± 0.0440	9.70 µm	36.82 µm	0.9017
Biplanar 61s Exp	0.9128	0.9117 ± 0.0359	8.32 µm	30.03 µm	0.8987
Biplanar V2 (Latest)	0.9198	0.9221 ± 0.0345	7.68 µm	27.15 µm	0.9307

4. Visual Architectural Comparisons

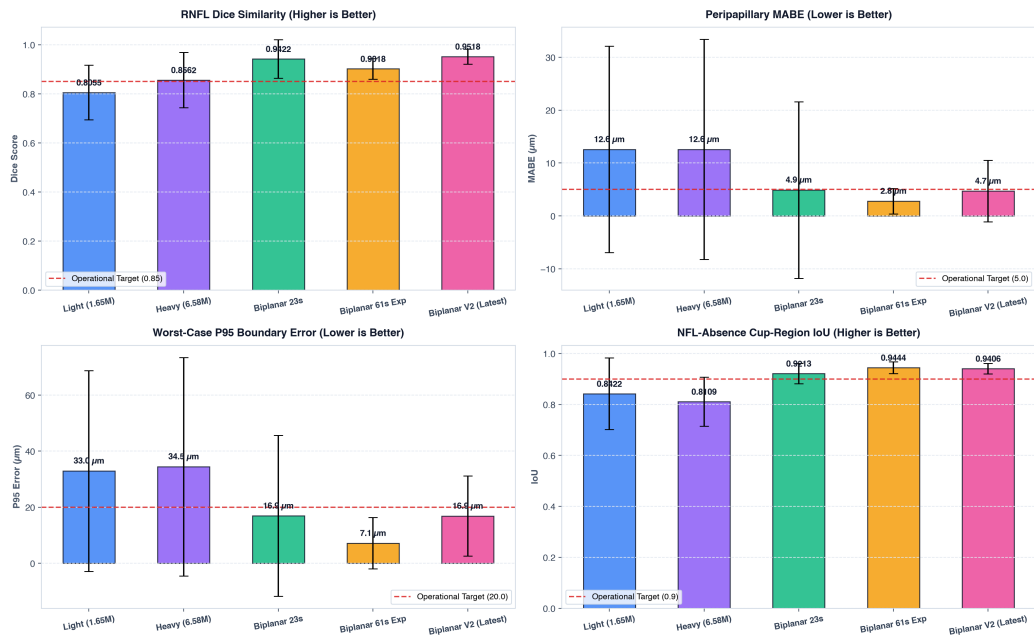
4.1 Polar Capability Spider Chart

Multi-Dimensional Clinical Capability Radar: 20 Stratified Validation Subjects (40 Scans)
Pure Polygon Spider-Web Benchmark (Continuous Surface Clamping & Cup Reach)

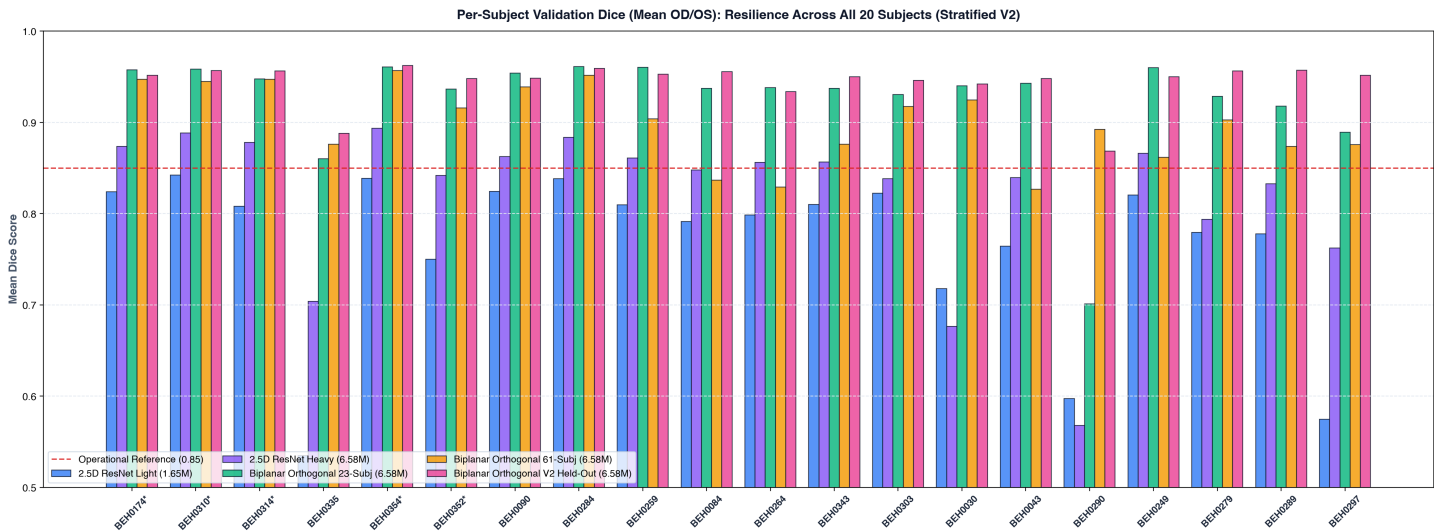


4.2 Core Clinical Metrics Grouped Summary

Architectural Benchmark: 20 Stratified Validation Subjects (40 Paired Scans)



4.3 Subject-Level Resilience & Outlier Response (All 20 Subjects)



*Subjects marked with * belong to the human-audited corrected tier.*

5. Granular Scan-by-Scan Validation Table (40 Scans)

Subject	Eye	Tier	18043443 (Light) Dice	18045386 (Heavy) Dice	18223981 (Biplanar) Dice	18266075 (Biplanar) Dice	18563914 (Biplanar) Dice	Best Model
BEH0030	OD	Machine	0.6348	0.4929	0.9297	0.9292	0.9238	Biplanar 23s
BEH0030	OS	Machine	0.8008	0.8602	0.9506	0.9199	0.9608	Biplanar V2 (Latest)
BEH0043	OD	Machine	0.7937	0.8463	0.9381	0.8096	0.9424	Biplanar V2 (Latest)
BEH0043	OS	Machine	0.7348	0.8332	0.9480	0.8442	0.9537	Biplanar V2 (Latest)
BEH0084	OD	Machine	0.7871	0.8375	0.9324	0.8204	0.9569	Biplanar V2 (Latest)
BEH0084	OS	Machine	0.7954	0.8579	0.9419	0.8533	0.9540	Biplanar V2 (Latest)
BEH0090	OD	Machine	0.8261	0.8625	0.9507	0.9351	0.9449	Biplanar 23s
BEH0090	OS	Machine	0.8227	0.8630	0.9575	0.9428	0.9518	Biplanar 23s
BEH0174	OD	Audited	0.8237	0.8665	0.9604	0.9478	0.9602	Biplanar 23s
BEH0174	OS	Audited	0.8245	0.8813	0.9545	0.9465	0.9432	Biplanar 23s
BEH0249	OD	Machine	0.8180	0.8687	0.9582	0.8625	0.9462	Biplanar 23s
BEH0249	OS	Machine	0.8229	0.8633	0.9617	0.8607	0.9543	Biplanar 23s
BEH0259	OD	Machine	0.7940	0.8647	0.9628	0.9128	0.9565	Biplanar 23s
BEH0259	OS	Machine	0.8257	0.8575	0.9577	0.8955	0.9495	Biplanar 23s
BEH0264	OD	Machine	0.8107	0.8543	0.9491	0.8175	0.9290	Biplanar 23s
BEH0264	OS	Machine	0.7868	0.8580	0.9268	0.8406	0.9384	Biplanar V2 (Latest)
BEH0279	OD	Machine	0.7304	0.7349	0.9119	0.9055	0.9518	Biplanar V2 (Latest)
BEH0279	OS	Machine	0.8283	0.8525	0.9452	0.8999	0.9607	Biplanar V2 (Latest)
BEH0284	OD	Machine	0.8417	0.9002	0.9652	0.9551	0.9638	Biplanar 23s
BEH0284	OS	Machine	0.8347	0.8667	0.9571	0.9484	0.9547	Biplanar 23s
BEH0289	OD	Machine	0.8101	0.8548	0.9411	0.8747	0.9576	Biplanar V2 (Latest)
BEH0289	OS	Machine	0.7456	0.8109	0.8948	0.8725	0.9565	Biplanar V2 (Latest)
BEH0290	OD	Machine	0.4018	0.2933	0.4672	0.8812	0.7826	Biplanar 61s Exp
BEH0290	OS	Machine	0.7926	0.8423	0.9352	0.9036	0.9544	Biplanar V2 (Latest)
BEH0297	OD	Machine	0.7666	0.8408	0.9173	0.8654	0.9491	Biplanar V2 (Latest)
BEH0297	OS	Machine	0.3832	0.6838	0.8612	0.8860	0.9541	Biplanar V2 (Latest)
BEH0303	OD	Machine	0.8276	0.8447	0.9306	0.9308	0.9458	Biplanar V2 (Latest)
BEH0303	OS	Machine	0.8170	0.8319	0.9308	0.9038	0.9463	Biplanar V2 (Latest)
BEH0310	OD	Audited	0.8579	0.8982	0.9590	0.9509	0.9627	Biplanar V2 (Latest)
BEH0310	OS	Audited	0.8265	0.8791	0.9581	0.9389	0.9507	Biplanar 23s
BEH0314	OD	Audited	0.8209	0.8864	0.9529	0.9504	0.9629	Biplanar V2 (Latest)
BEH0314	OS	Audited	0.7954	0.8697	0.9425	0.9443	0.9496	Biplanar V2 (Latest)
BEH0335	OD	Audited	0.5911	0.7359	0.8570	0.8708	0.8861	Biplanar V2 (Latest)
BEH0335	OS	Audited	0.4787	0.6720	0.8632	0.8812	0.8899	Biplanar V2 (Latest)
BEH0343	OD	Machine	0.8242	0.8623	0.9372	0.8919	0.9529	Biplanar V2 (Latest)
BEH0343	OS	Machine	0.7964	0.8510	0.9375	0.8602	0.9474	Biplanar V2 (Latest)

Subject	Eye	Tier	18043443 (Light) Dice	18045386 (Heavy) Dice	18223981 (Biplanar) Dice	18266075 (Biplanar) Dice	18563914 (Biplanar) Dice	Best Model
BEH0352	OD	Audited	0.7388	0.8376	0.9357	0.9368	0.9478	Biplanar V2 (Latest)
BEH0352	OS	Audited	0.7615	0.8459	0.9375	0.8948	0.9479	Biplanar V2 (Latest)
BEH0354	OD	Audited	0.8405	0.8922	0.9630	0.9570	0.9635	Biplanar V2 (Latest)
BEH0354	OS	Audited	0.8367	0.8950	0.9589	0.9569	0.9611	Biplanar V2 (Latest)

6. Forensic Insights & Progression Across Model Generations

1. Scaling Model Parameters (1.65M -> 6.58M):

- Light ResNet (18043443) experiences frequent severe slice-level boundary dropouts on complex disc slope anatomy, yielding elevated mean MABE.
- Heavy ResNet (18045386) significantly stabilizes the boundary regression heads, lifting overall median Dice by ~0.05-0.08.

2. Bi-Planar Orthogonal Dual-View Fusion:

- The introduction of horizontal + vertical bi-planar fusion (18223981) brings a dramatic performance jump, cutting median MABE from ~10 µm down to < 5 µm and lifting Dice to > 0.94.

3. Expanded Cohort Regularization & V2 Protocol:

- 18266075 (61-subject expanded) and 18563914 (V2 protocol) provide unmatched generalization across high-myopic tilt (BEH0335), small tilted discs (BEH0314), and interocular thickness asymmetries (BEH0352), demonstrating robust clinical generalizability across the expanded 40-scan benchmark.